

Thomison Le

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.S. in Mechanical Engineering - GPA: 3.62/4.00

Expected May 2029

- Relevant coursework: Linear Algebra and Differential Equations, Foundations of Data Science, Computer Programming and Numerical Methods, Multivariable Calculus, Physics for Scientists and Engineers

SKILLS

Analysis and Modeling: Python, MATLAB, CAD, Altium, Figma, C++, ArcGIS, Google Sheets, Microsoft Excel

Fabrication and Machining: OMAX Waterjet, Tormach CNC, 3D Printing, Soldering/Wire Harnessing, Woodshop

EXPERIENCE & ACTIVITIES

Undergraduate Research Intern - Model Predictive Control Lab | Berkeley

May 2026 - Present •

- Diagnosed MCU-PC failure on a **\$50k+** Ridgeback robot; replaced CMOS battery and configured BIOS to restore use.
- Validating asymmetric control algorithms for human-robot navigation across **200+** Monte-Carlo simulations.
- Building ROS pipelines to deploy **2D** dynamic game models for safe, cooperative human-robot transport.

Avionics & Manufacturing Engineer - Space Enterprise at Berkeley | Berkeley

September 2025 - Present

- Engineered a Python (**Pandas/Numpy**) telemetry pipeline to synchronize 100Hz multi-sensor flight logs, applying numerical double integration to IMU acceleration data to model displacement and evaluate GPS/Barometer drift.
- Implemented quaternion math to translate IMU data into a world frame from a fixed body frame.
- Fabricated flight-critical avionics assemblies, using precision wire harnessing and soldering to ensure rocket integrity.
- Utilized abrasive waterjet machining to manufacture flight-critical rocket components with precision tolerances.

Data Analysis Intern - The Energy Coalition | San Jose

June 2024 - August 2024

- Developed an interactive ArcGIS StoryMap to visualize local energy and water usage trends, presenting case studies on urban sustainability and climate resilience for San Jose, educating **100+ users**.
- Conducted comprehensive energy and water audits, utilizing Google Sheets to model consumption rates and utility costs.
- Synthesized complex datasets into Comparative Energy Analysis reports to identify inefficient infrastructure and recommend improved alternatives, resulting in a projected **74%** reduction in energy costs.

Scholar - Summer Math and Science Honors Academy | Berkeley

June 2024

- Utilized Figma and Python to prototype "FirstGenAdvisor", a web app dedicated to benefiting underprivileged students, reaching the semifinals across a nationwide competition consisting of **500+ scholars**.
- Rapidly upskilled in Figma and Python to collaborate with peers and graduate student mentors to execute project goals.

PROJECTS

Currency Detector

June 2026 - Present

- Trained 10+ custom YOLOv26 object detection models to classify US banknotes, ultimately achieving a mAP50 of **70.7%** across 10 classes on a dataset that comprised 10,000+ images.
- Designed an interactive web application to run client-side machine learning inference directly inside a user's browser.

Natural Language Movie Classifier

April 2026

- Built a k-Nearest Neighbors NLP model in Python (**Pandas**) to classify movie genres from screenplays.
- Processed **300+** movie scripts, extracting **10+** optimally selected word frequency features, tuning the algorithm to k=15.
- Achieved a **70%** predictive accuracy on unseen test data using an optimized 85/15 train-test split pipeline.

Autonomous Mobile Robot

December 2025

- Developed a simulated physics-based testing environment in Python (**PyBullet**) to validate mechanical designs against real-world factors like friction, gravity, and collision impact, reducing the need for physical iteration.
- Implemented a logical navigation protocol (A*), engineering specific recovery behaviors for edge cases where the system became physically stuck, reducing entrapment by **95%**.
- Designed a modular mapping system to manage complex grid occupancy state data with strict logic error tolerance.

LEADERSHIP & VOLUNTEER WORK

Youth Partner & Research Assistant - InTandem Fellowship | Remote

May 2024 - Present

- Represented youth voice to companies and organizations such as Meta and OpenAI in areas like app optimization, mental health, and education in online seminars, app demos, and interviews.

Mentor - Berkeley Engineers and Mentors | Berkeley

February 2026 - Present

- Educated 15+ mentees (TK, Kindergarten) weekly on scientific and engineering topics through physical hands-on learning.